

Finally, literature on the differences between groups of authors formed on the basis of productivity points to permanent stratification linked to top journals. The first author has described (Kwiek, 2021) the concentration of authorship in elite journals and the advantage of frequent publishers have over infrequent ones (in top journals in higher education research in the past 30 years). The empirical results strengthen a view of top journals as a publishing segment with high entry barriers and strong reproduction mechanisms. Romero-Silva and colleagues (2025) discussed the dominance of leading business schools in top journals, and Balzer and Benlahlou (2025) analyzed prestige bias in citations, which is consistent with the argument that cumulative advantage is embedded in prestige channels.

There are two hypotheses explaining the grip of top journals on individual scientists (Fochler et al., 2016; Lindahl, 2018), institutions (Mouritzen & Opstrup, 2020), and national systems (Franzoni et al., 2011). One is the prestige-maximization model (see Kwiek, 2021), located within broader theories of academic capitalism and resource dependence (Slaughter & Leslie, 1997), which explains how universities and scientists work together for the similar goal of prestige maximization. The other is principal-agent theory, which explains how publishing in prestigious journals aligns the interests of individual scientists (as agents) with those of their institutions and research-sponsoring organizations, including national governments (as principals; Braun & Guston, 2003; Kivistö, 2008; van der Meulen, 1998).

According to the prestige-maximization model, research universities, their departments, and individual scientists act as “prestige maximizers” (Melguizo & Strober, 2007; Taylor et al., 2016). Prestige received from publications in top journals can be converted into research grants; and institutions, departments, and scientists modify their behavior, including publishing patterns, accordingly, in competing for external resources (Taylor et al., 2016). Publishing in elite journals became especially important following the development of research assessment systems (van Dalen & Henkens, 2005; Whitley & Glaser, 2007). Research funding from prestigious granting agencies is another avenue of prestige maximization.

Principal-agent theory can be applied to better understand the role of top journals for governments. The theory was initially used in studies of corporations (Pratt & Zeckhauser, 1985), but then it was applied to science sectors (Braun & Guston, 2003; Kivistö, 2008; van der Meulen, 1998). In the relationship between research universities and individual scientists as agents and governments as principals, publication in top journals is a key indicator of institutional productivity and a critical component in competition for funding, including the competition within various national excellence initiatives. (The Polish version of a national excellence initiative, the IDUB Program of 2020–2026, was built entirely on Scopus- and WoS-defined indicators of top journals).

Generally, it is almost impossible for the principal to understand the agent’s products (Braun & Guston, 2003, pp. 303–304), that is, research publications, or to assess their impact. At the same time, the principal must ensure that academics produce high-quality research. The label of publication in top journals enables principals at all levels (national, institutional, and departmental) to defend their distribution of rewards in a rational manner. Top journals can also easily serve as a common performance metric in most disciplines (Gomez-Mejia & Balkin, 1992, p. 925).

Principal-agent theory offers a useful way to understand the appeal of top journals to both principals and agents (for whom it is easier to demonstrate their own academic success). Instead of reading thousands of papers in the way peer review suggests, the theory reduces the burden of quality assessment. The number of papers is radically reduced if only publications in top journals are

reported (and they can be easily traced across disciplines, academic units, institutions, and countries).

The integration of the classical literature on inequalities in science, the literature on visibility and prestige, and studies on measures of productivity, publishing elites, and top journals leads to a conclusion that research has often treated the relationship between high productivity and high journal prestige as obvious and unidirectional. Meanwhile, contemporary science is ever more strongly organized by hierarchies of journals that may act as mechanisms of elite selection and reproduction.

Our aim is not to confirm that top performers publish much more than others (as we have done; see Kwiek, 2016, 2018; Kwiek & Roszka, 2024) but to confirm that differences between top performers and other scientists are qualitative in nature: they are concentrated mostly in the highest ranks of journals. This study interprets these patterns as durable stratification mechanisms in which selective access to top journals is a key source of long-term advantage.

## **2.1. Research questions and hypotheses**

The aims of this study are to understand the relationships between publication productivity and access to prestigious journals and to assess the extent to which publishing in top journals constitutes a stratification mechanism of the publishing elite. In contrast to approaches that focus exclusively on publication numbers, in this research we assume that productivity and the prestige of publication channels are related in a nonlinear and complementary manner, and access to top journals may serve as a selective career gate. In this context, the following research questions and hypotheses were formulated.

### **2.1.1. Research questions**

RQ1. Do top performers publish in top journals (90–99 Scopus journal percentile ranks) to a disproportionately higher extent than other scientists, regardless of discipline or period?

RQ2. Is the observed stratification between top performers and other scientists qualitative in nature – that is, do the differences between the two groups emerge primarily in the highest segments of the journal prestige hierarchy?

RQ3. How does the concentration of publications in top journals translate into the concentration of productivity measured by a journal prestige-normalized indicator?

RQ4. Is entry into the group of top performers more strongly associated with publication intensity itself or rather with a combination of a high number of publications and high quality of publication channels?

On the basis of these research questions, the following research hypotheses were formulated.

### **2.1.2. Research hypotheses**

H1 (*Prestige selection*)

Top performers achieve a significantly higher share of publications in top journals (journals located in the 90–99 Scopus percentile ranks) than other scientists in all disciplines and periods studied.

## H2 (*Qualitative stratification*)

The differences between top performers and the remaining scientists emerge primarily in the highest journal prestige deciles, while in the lower and middle segments of the journal hierarchy, the publication profiles of the classes are relatively similar.

## H3 (*Complementarity of quantity and quality*)

The effect of publishing intensity on the probability of membership in the class of top performers increases with the quality of publication channels, indicating a complementary – rather than a supplementary – effect of quantity and quality of scientific production.

## H4 (*Structural character of stratification*)

The stratification of access to top journals is structural: it persists over time (30 years) despite increases in publication volume, and it occurs in all disciplines, albeit with varying intensity.

# 3. Dataset, sample, and methods

## 3.1. Dataset

We used a Scopus dataset with authors and publications extracted in November 2022–January 2023. We also used supplementary data provided by the International Center for the Study of Research (ICSR Lab) run by Elsevier, the owner of the Scopus dataset. We used the complete, in terms of coverage, dataset from 1992 to 2022. This period corresponds to the availability of consistent Scopus coverage for Polish-affiliated authors and allows for the analysis of long-term structural patterns rather than short-term fluctuations. We used data from all Polish research-involved institutions (in all sectors, 343 in total) though the vast majority and authors and publications come from the higher education sector. We used separate affiliations in each of the six five-year periods. Our approach was to select the dominating affiliation in the publications' bylines in the period. However, the only institutional variable used was IDUB vs. all other institutions (the IBUB variable characterizing 10 selected research-intensive universities, all participants in the Polish national research excellence program). This binary distinction was adopted to capture broad differences in research intensity of Polish institutions. No differences are made among non-IDUB institutions, so the class includes all remaining universities, institutes of the Polish Academy of Sciences, research institutes, and corporate bodies. Mobility between institutions, as captured by changing affiliations, is marginal in Poland and is noted especially for young scientists who are employed on research grants. As a result, affiliation-based measurement error is unlikely to substantially affect individual productivity estimates within periods.

Scientists assigned by Scopus to any of the 343 institutions were regarded as Polish scientists. We also treated as scientists co-authors who were graduate students (very rare cases), as they were listed as Scopus authors with individual IDs. Individuals (rather than publications) were the unit of analyses. This approach allows productivity to be interpreted as an individual-level characteristic rather than an artifact of publication-level aggregation. Consequently, different or multiple affiliations in a period do not change the overall productivity level in that period. The institutional research intensity variable was used only in our model approach.

We had complete bibliometric data about individual scientists: full names, information about affiliations (from publication bylines), disciplines (the modal value: to which did they refer most frequently in all their research articles and papers in conference proceedings indexed in Scopus),

and their unique Scopus ID. The modal approach reduces the influence of occasional cross-disciplinary publications and reflects the dominant orientation of a scientist's research activity. We also had the year of each scientist's first publication (indicating academic age), experience at the time of publishing any article or at the end of each sex-year period. Academic age thus captures cumulative publishing exposure rather than biological age or formal career stage. To define gender, we used the gender-defining tool *genderize.io*, with the accuracy threshold of 0.85.

We conducted thorough data cleaning, removing from the analysis 6,885 cases for which gender could not be established, 14,431 cases for which the year of the first publication could not be established, and 49,939 cases for which the dominant discipline could not be established. Finally, at the publication level, publications with more than 100 authors were removed from the dataset. The areas of study were restricted to 15 STEMM disciplines (science, technology, engineering, mathematics, and medicine according to the All Science Journal Classification [ASJC] used in Scopus). Finally, the dataset included more than 144,000 scientists and more than 433,000 publications.

### 3.2. Sample description

The analysis is based on a full-population dataset (all internationally visible scientists) comprising  $N = 144,314$  Polish researchers publishing between 1992 and 2021 whose output consisted of a total of  $N_{art} = 433,546$  unique research articles indexed in Scopus. The data were aggregated into five six-year analytical periods, which allowed us to capture long-term structural changes in the science system, rather than merely short-term fluctuations. The authors came from all science sectors (higher education, research institutes, and the private sector) and could be employed full-time or part-time. For productivity calculations, we used only research articles (minimum threshold: one article in the whole period).

In all periods, an increase in the size of each academic age group is observed, with the fastest growth in the youngest cohort. The number of articles steadily increased in successive analytical periods – from 28,241 publications in 1992–1997 to 156,611 in 2016–2021. Table 1 shows the distribution of researchers in each subperiod: some scientists published in one period, others in all of them. Therefore, the total of scientists for all periods (264,283) is much higher than the number of unique scientists (144,314).

**Table 1.** Sample description – authors in the six periods

| Feature    | Category | 1992–1997 | 1998–2003 | 2004–2009 | 2010–2015 | 2016–2021 | Total   |
|------------|----------|-----------|-----------|-----------|-----------|-----------|---------|
| Gender     | female   | 6,935     | 12,979    | 22,509    | 34,545    | 46,417    | 123,385 |
|            | male     | 12,299    | 18,656    | 26,894    | 37,749    | 45,300    | 140,898 |
| Discipline | AGRI     | 1,338     | 2,508     | 4,328     | 7,158     | 9,223     | 24,555  |
|            | BIO      | 2,162     | 3,428     | 4,947     | 7,018     | 8,835     | 26,390  |
|            | CHEM     | 2,767     | 4,316     | 5,714     | 7,453     | 8,157     | 28,407  |
|            | CHEMENG  | 167       | 312       | 463       | 473       | 551       | 1,966   |
|            | COMP     | 154       | 282       | 657       | 1,413     | 1,757     | 4,263   |
|            | EARTH    | 691       | 1,292     | 1,707     | 2,510     | 3,282     | 9,482   |
|            | ENER     | 34        | 80        | 215       | 537       | 1,029     | 1,895   |
|            | ENG      | 811       | 1,342     | 3,051     | 5,424     | 7,788     | 18,416  |
|            | ENVI     | 322       | 678       | 1,305     | 2,465     | 3,831     | 8,601   |
|            | MATER    | 752       | 1,151     | 2,105     | 3,624     | 4,963     | 12,595  |

|                     |       |        |        |        |        |        |         |
|---------------------|-------|--------|--------|--------|--------|--------|---------|
|                     | MATH  | 856    | 1,207  | 1,633  | 2,135  | 2,208  | 8,039   |
|                     | MED   | 5,606  | 10,381 | 17,771 | 25,374 | 32,672 | 91,804  |
|                     | NEURO | 202    | 335    | 419    | 619    | 871    | 2,446   |
|                     | PHARM | 305    | 331    | 361    | 470    | 525    | 1,992   |
|                     | PHYS  | 3,067  | 3,992  | 4,727  | 5,621  | 6,025  | 23,432  |
| Academic age groups | 0–9   | 11,640 | 20,386 | 31,223 | 44,186 | 52,981 | 160,416 |
|                     | 10–19 | 4,591  | 5,836  | 10,136 | 17,183 | 22,304 | 60,050  |
|                     | 20–29 | 2,552  | 4,039  | 4,943  | 5,973  | 10,170 | 27,677  |
|                     | 30+   | 451    | 1,374  | 3,101  | 4,952  | 6,262  | 16,140  |
| Total               |       | 19,234 | 31,635 | 49,403 | 72,294 | 91,717 | 264,283 |

Note: The number of scientists is higher than the number of unique scientists. Each scientist can belong to 1–5 periods based on their publishing history.

**Table 2.** Sample description – articles

| Period    | Number of articles |
|-----------|--------------------|
| 1992–1997 | 28,241             |
| 1998–2003 | 49,869             |
| 2004–2009 | 78,547             |
| 2010–2015 | 120,278            |
| 2016–2021 | 156,611            |
| Total     | 433,546            |

### 3.3. Methods

We used bibliometric publication and citation metadata from Scopus. As in recent research (Kwiek & Roszka, 2024, 2025), our general approach has four major features. It is longitudinal (we trace Polish scientists over subsequent six-year periods, spanning the thirty years from 1992 to 2021); it is relative (we allocate scientists to the two classes of top performers and the remaining scientists: the top 10% of scientists in terms of productivity within disciplines and within periods, and the remaining 90%); it is classificatory (we allocate all scientists to one of the two classes based on their productivity rank); and it is nonlinear and journal prestige-normalized (we recognize that linear and non-normalized approaches to productivity do not reflect the differences between publishing in top-tier journals and publishing in bottom-tier journals strongly enough, as global academic journals are highly stratified).

In our approach to measuring productivity, we amplified the distance between elite journals and mid-tier journals, thereby more accurately modeling the steep stratification of the publication market. We linked individual publications to journals and their global Scopus percentile ranks (0–99) using our dataset; we were not able to link our publications to their publications over time. Working with article-level citation metrics would be preferable; however, in the Polish context, working with journal percentile ranks is very close to working with publication points routinely used by Polish scientists in the past three decades. The basic assumption of Polish research evaluation systems has been that academic journals are not equal, which is expressed in the number of points attached to them (currently in the 20–200 range). The correlation of Scopus journal lists and national lists of journals (with the points assigned), for the STEM disciplines in this study, is very high.

We constructed individual publication portfolios separately for periods. The portfolios included all Scopus-derived metadata on publications: journal metadata and publication metadata. We chose all